Date: _____

Ghazi University

CS and IT department

Subject: programming fundamentals

Topic: PFLAB-2

Section: BSCS-C

Submitted by: Misbah Rehman

Roll No: 27

Submitted To: Sir Asad Majeed

Date: 09-10-2026

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Date: _____

TASK-1:

Type and solve the following program in visual studio Run the program and observe their output.

⇒ Input

```
# include <iostream>
using namespace std;
  int main( ) {
    char letter;
    letter = 'A';
cout << letter<< '\n';
    letter = 'B';
 cout << letter<< '\n';
  cin >> letter;
     return 0;
         }
```

output

```
A
B
```

⇒ Input

```
# include <iostream>
using namespace std;
```

output

```
A$
B6
```

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return 0;
}

TASK-5: Calculate Area of circle

Input

```
#include<iostream>
using namespace std;
int main ( ){
float r, area circle;
Const float pI = 3.14159;
Cout << "Enter radius:";
  Cin >> r;
area circle = pI * r * r;
Cout<<"Area of circle= "<< areacircle<<endl;
   return 0;
}
```

output

```
Enter radiu: 5
Area of
circle: 78.54
```

⇒calculate area of Rectangle

Input

```
#include<iostream>
using namespace std;
 int main ( ){
float length, width;
Cout<<"Enter length & width!";
 Cin >>length >> width;
 cout<<" Area of Rectangle = "<<l*w<<endl;
 return 0;
```

output

```
enter length and w
4, 6;
Area of Rectan
24
```

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=>Calculate Area of Triangle

Input

```
# include <iostream>
using namespace std;
int main ( ) {
float base, height;
cout << "enter base and height: ";
cin >> base >> height;
cout << "Area of triangle = " << base * height * 0.5 << endl;
    return 0;
}
```

output

```
enter base and height:
4 6;
Area of triangle:
12
```

TASK-6: Total purchase a customer in a store is purchasing five items. The price of five items are;

Input

```
#include <iostream>
using namespace std;
int main( ) {
double P1=12.9$, P2=24.9$, P3=6.9$, P4=14.9$, P5=3.9$;
double subtotal, salesTax, total;
Subtotal = P1 + P2 + P3 + P4 + P5;
Sales TAX = subtotal * 0.06;
total = Subtotal + Sales Tax;
cout << "price1: $" << price1 << endl;
cout << price2: $" << price2 << endl;
cout << price3: $" << price3 << endl;
```

output

```
P1 = $ 12.9 S
P2 = $ 24.9 S
Price 3 = $ 6.9 S
price 4 = $ 14.9 S
Price S = $ 3.9 S
Subtotal = $ 63.7
```

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```
cout<<"price 4: "$"<<price4<<endl;
cout<<"price 5: $"<<price5<<endl;
cout<<"Subtotal:$"<<Subtotal<<endl;
cout<<Sales TAX(6%):$"<<Sales tax<<endl;
cout<<"Total: $"<<total<<endl;
    return 0;
    }
```

Sales tax (6%) = $3.82
Total = $67.575

BONUS TASK: Break number

```
#include<iostream>
using namespace std;
int main ( ) {
int num ;
cout<<"enter five-digit integer: ";
cin >> num ;
 int d1 = num /10000;
 int d2 = (num/1000 )% 10;
int d3= (num /100) % 10;
int d4 = (num/10 ) % 10;
int d5 = num % 10;
cout<<d1<<" "<<d2<<" "<<d3<<" "<<d4<<" "<<d5<<" "<<endl;
    return 0;
    }
```

output

```
enter five-digit integer = 6000
```

Date:

HOME TASK - 1: Swap Variable

```
#include <iostream>
using namespace std;
 int main( ) {
int var1, var2, temp;
cout <<"enter var1 and var2: ";
cin >> var1 >> var2;
cout <<" before swap: var1 = 10 \n";
cout << before swap: var2 = 20 \n";
 temp = var1;
  var1 = var2;
   var2 = temp;
cout <<"after swap: var1 = 20 \n";
cout << "after swap: var2 = 10";
    return 0;
      }
```

Output:

```
enter var1 and var2: 10  20
before swap: var1 = 10
before swap: var2 = 20
after swap:: var1 = 20
after swap: var2 = 10
```

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Date:

HOME-TASK: 2: Small To capital

```
# include <iostream>
using namespace std;
int main ( ) {
Char Small;
Cout << "enter Small alphabet (a-z): ";
 Cin >> Small;
Char capital = Small - 32;
 Cout << " upper case: \n";
   return 0;
     }
```

output:

enter small alphabet (a-z): m
upper case:

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Project Execute Tools AStyle Window Help

[*] Untitled1.cpp

```
1   BONUS TASK: BREAK NUMBERS
2
3   # include <iostream>
4   using namespace std;
5
6   int main (){
7   int num;
8   cout<<" enter fivr digit integer: ";
9   cin>> num
10  int d1= num/10000;
11  int d2=( num/1000)%10;
12  int d3=( num/100)%10;
13  int d4=( num/10)%10;
14  int d5= num%10;
15  cout<<d1<<"  "<<d2<<"  "<<d3<<"  "d4<<"  "<<d5<<"  "<<endl;
16  return 0;
17  }
18
19
```

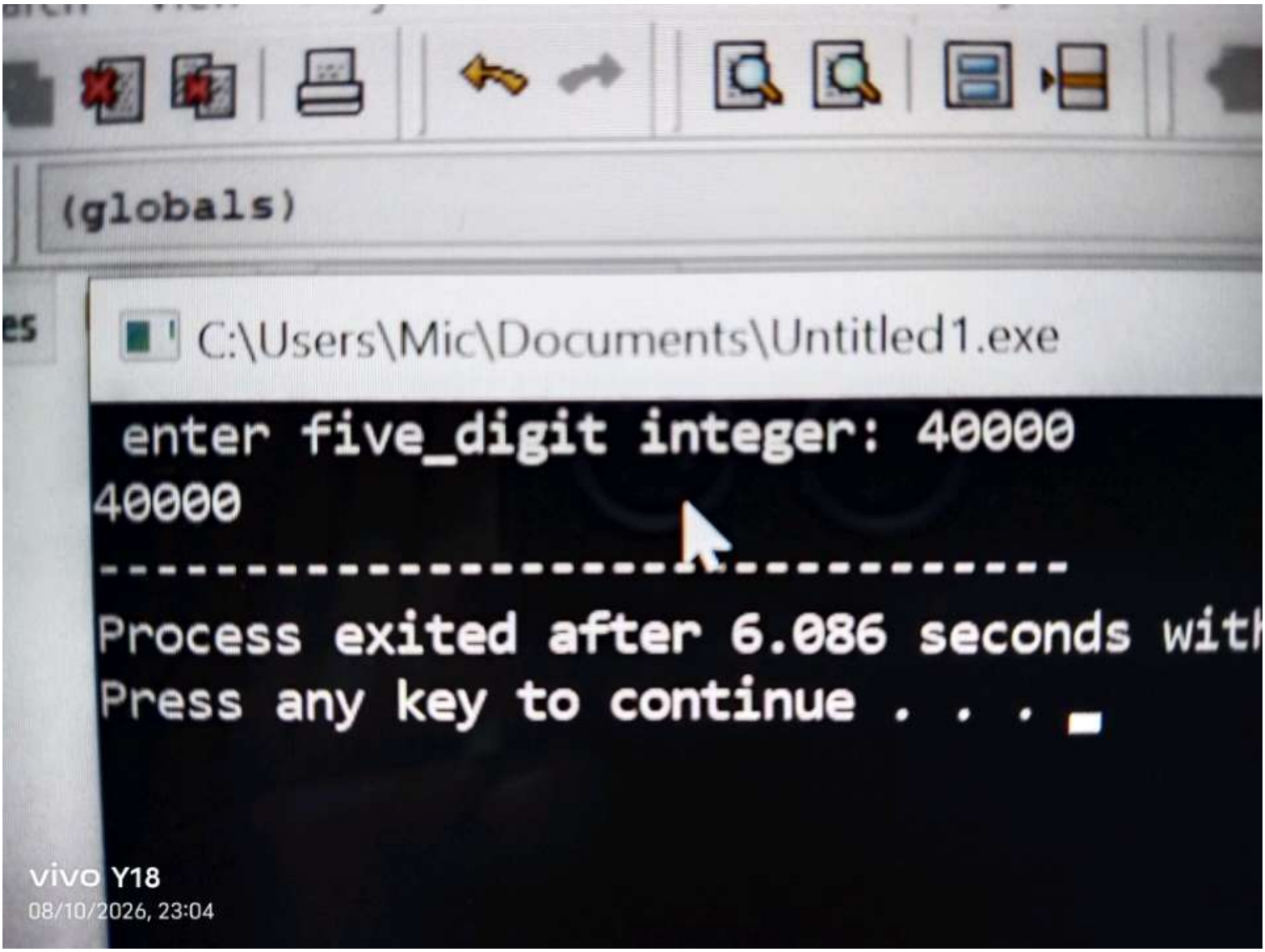
Resources Compile Log Debug Find Results

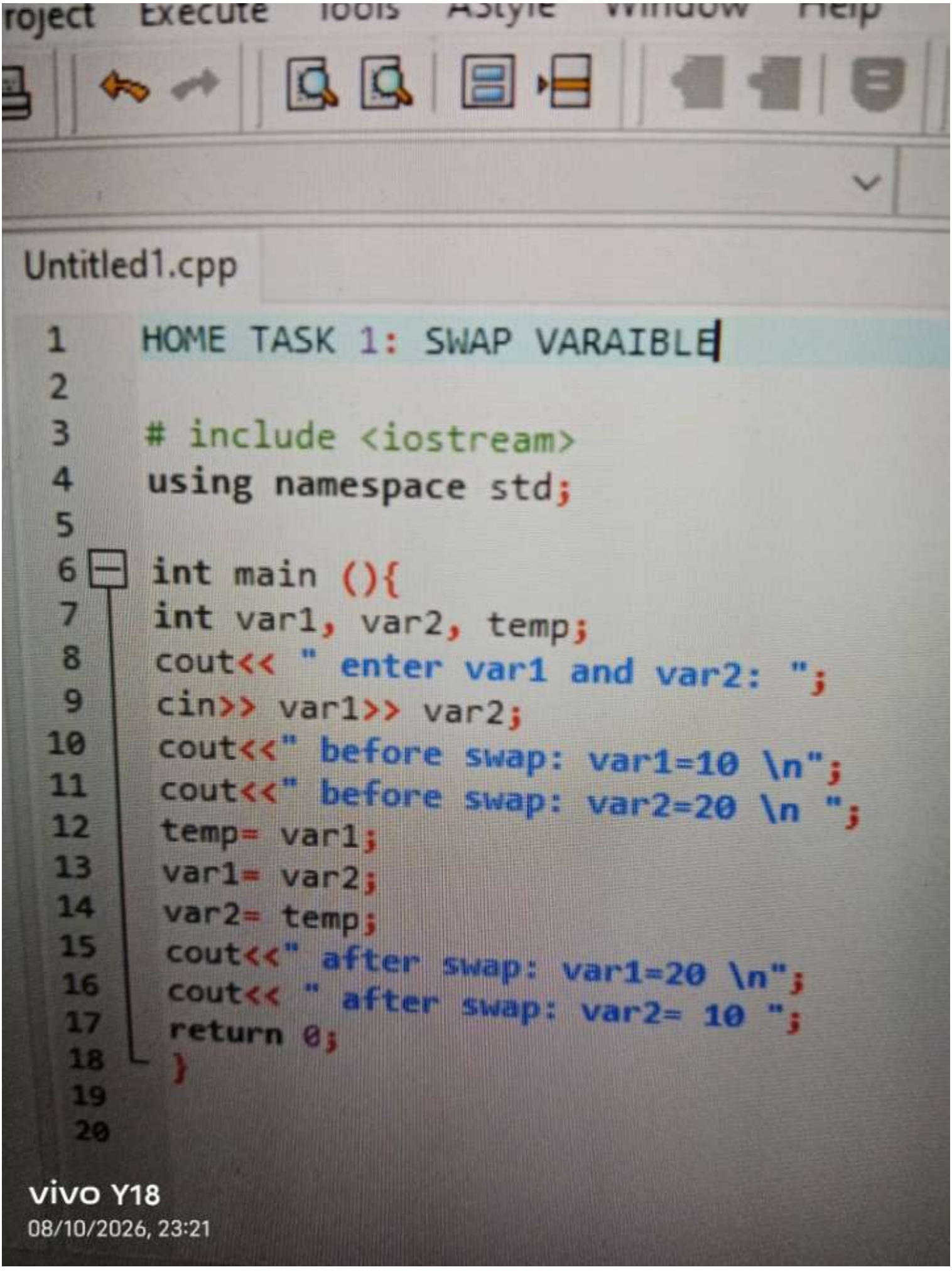
26 Sel: 0 Lines: 19 Length: 333 Insert Done parsing in 0.406

pe here to search

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(globals)
C:\Users\Mic\Documents\Untitled1.exe
enter five_digit integer: 40000
40000
Process exited after 6.086 seconds with
Press any key to continue . . .
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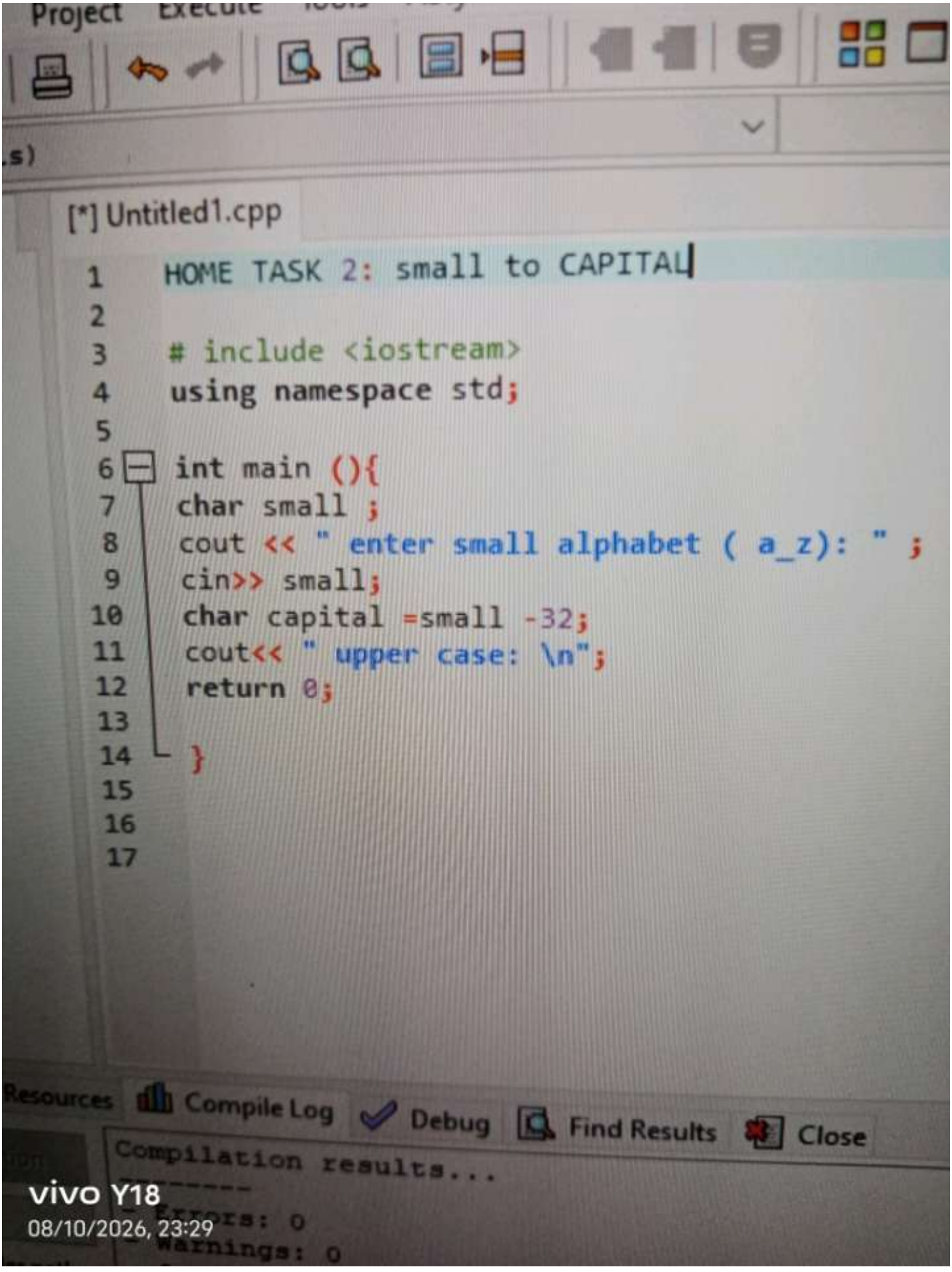

roject Execute Tools AStyle Window Help
Untitled1.cpp
1 HOME TASK 1: SWAP VARAIBLE
2
3 # include <iostream>
4 using namespace std;
5
6 int main (){
7 int var1, var2, temp;
8 cout<< " enter var1 and var2: ";
9 cin>> var1>> var2;
10 cout<<" before swap: var1=10 \n";
11 cout<<" before swap: var2=20 \n ";
12 temp= var1;
13 var1= var2;
14 var2= temp;
15 cout<<" after swap: var1=20 \n";
16 cout<< " after swap: var2= 10 ";
17 return 0;
18 }
19
20
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ug Untitled1.cpp

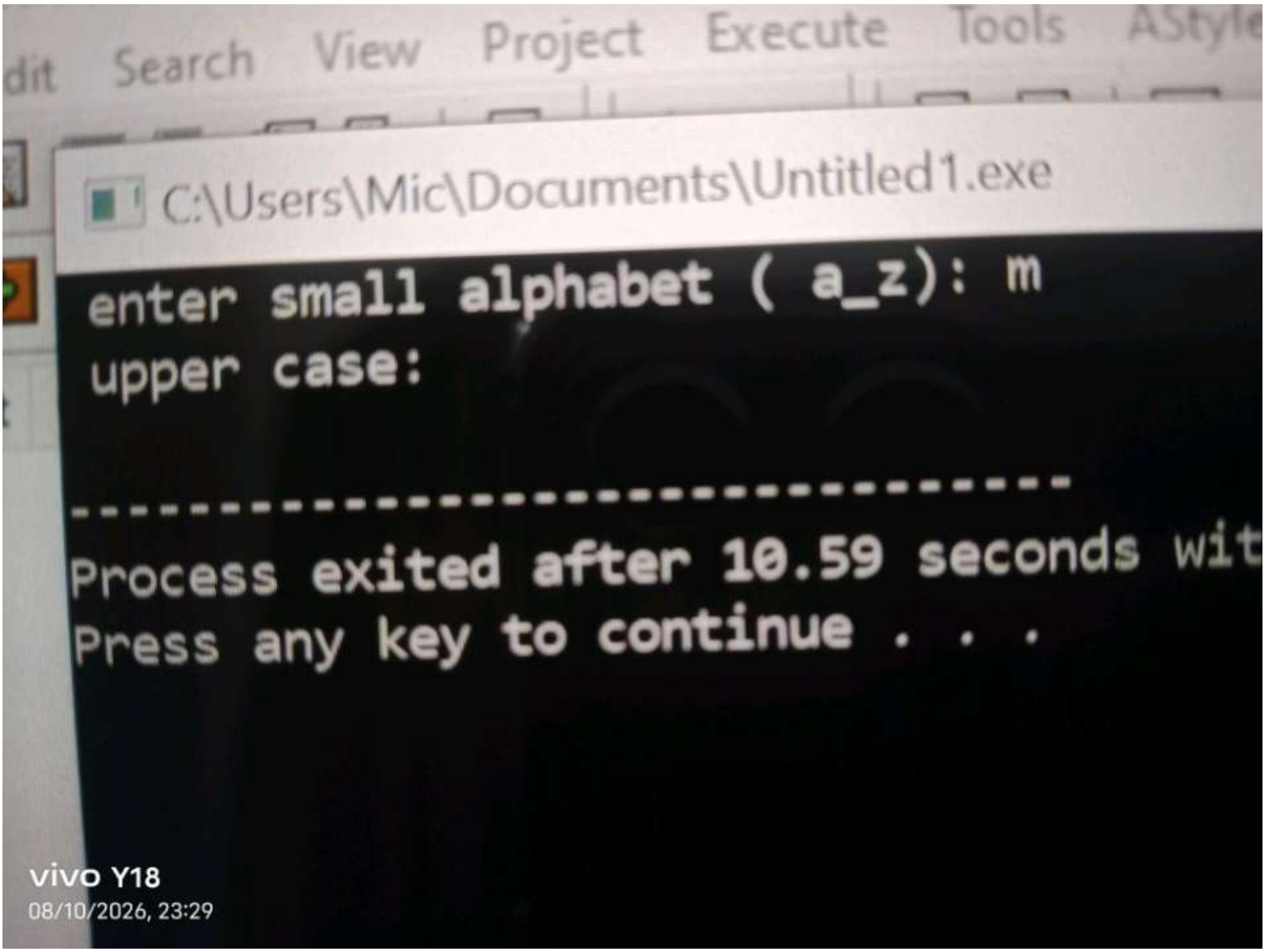
C:\Users\Mic\Documents\Untitled1.exe

```
enter var1 and var2: 10 20
before swap: var1=10
before swap: var2=20
 after swap: var1=20
after swap: var2= 10
--------------------------------
Process exited after 9.351 seconds wi
Press any key to continue . . .
```

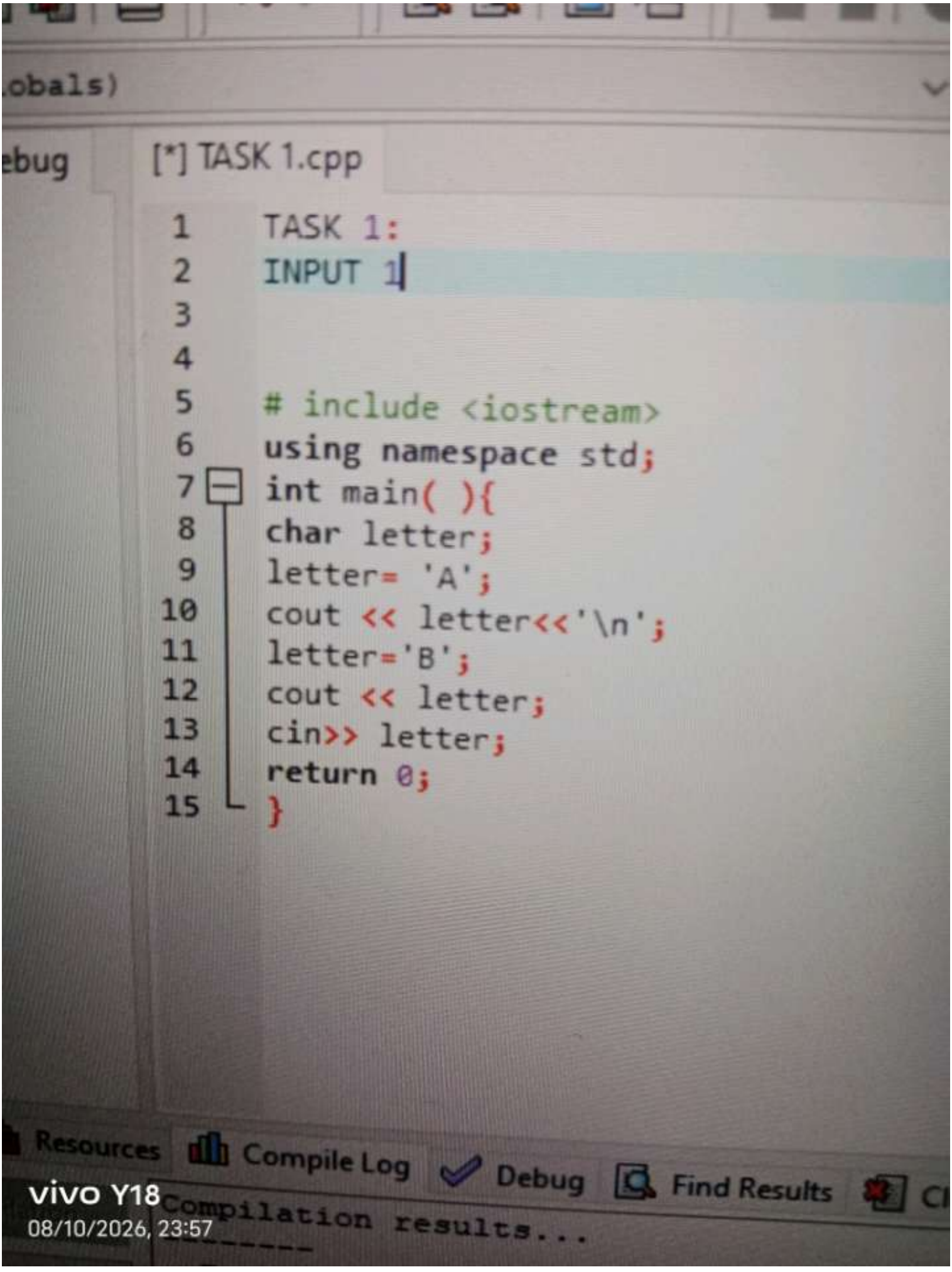
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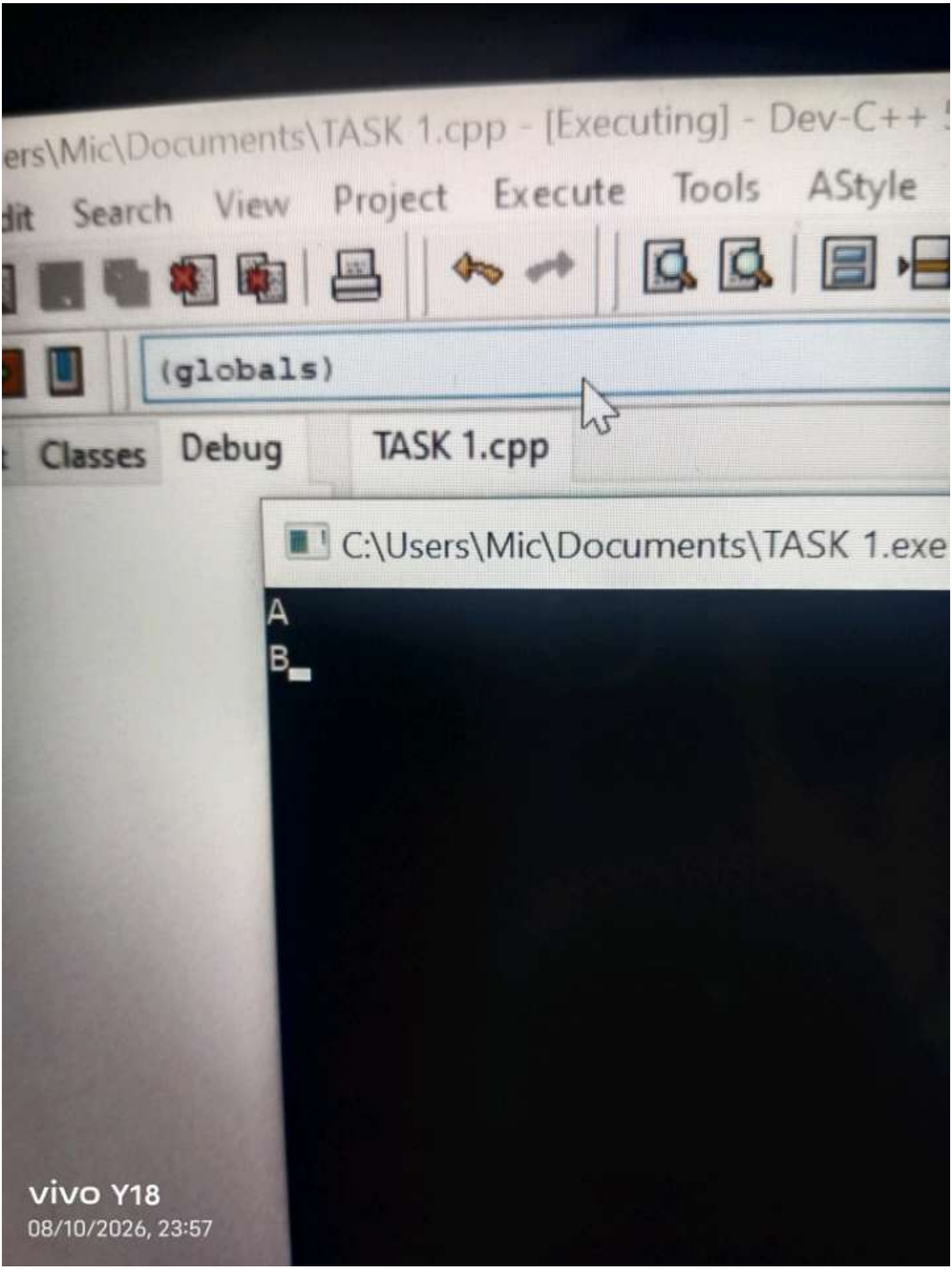


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dit Search View Project Execute Tools AStyle
C:\Users\Mic\Documents\Untitled1.exe
enter small alphabet (a_z): m
upper case:

Process exited after 10.59 seconds wit
Press any key to continue . . .
vivo Y18
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obals)
ebug
[*] TASK 1.cpp
1 TASK 1:
2 INPUT 1
3
4
5 # include <iostream>
6 using namespace std;
7 int main( ){
8 char letter;
9 letter= 'A';
10 cout << letter<<'\n';
11 letter='B';
12 cout << letter;
13 cin>> letter;
14 return 0;
15 }
Resources
Compile Log
Debug
Find Results
Compilation results...
vivo Y18
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ers\Mic\Documents\TASK 1.cpp - [Executing] - Dev-C++
dit Search View Project Execute Tools AStyle
(globals)
Classes Debug
TASK 1.cpp
C:\Users\Mic\Documents\TASK 1.exe
A
B
vivo Y18
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obals)

bug

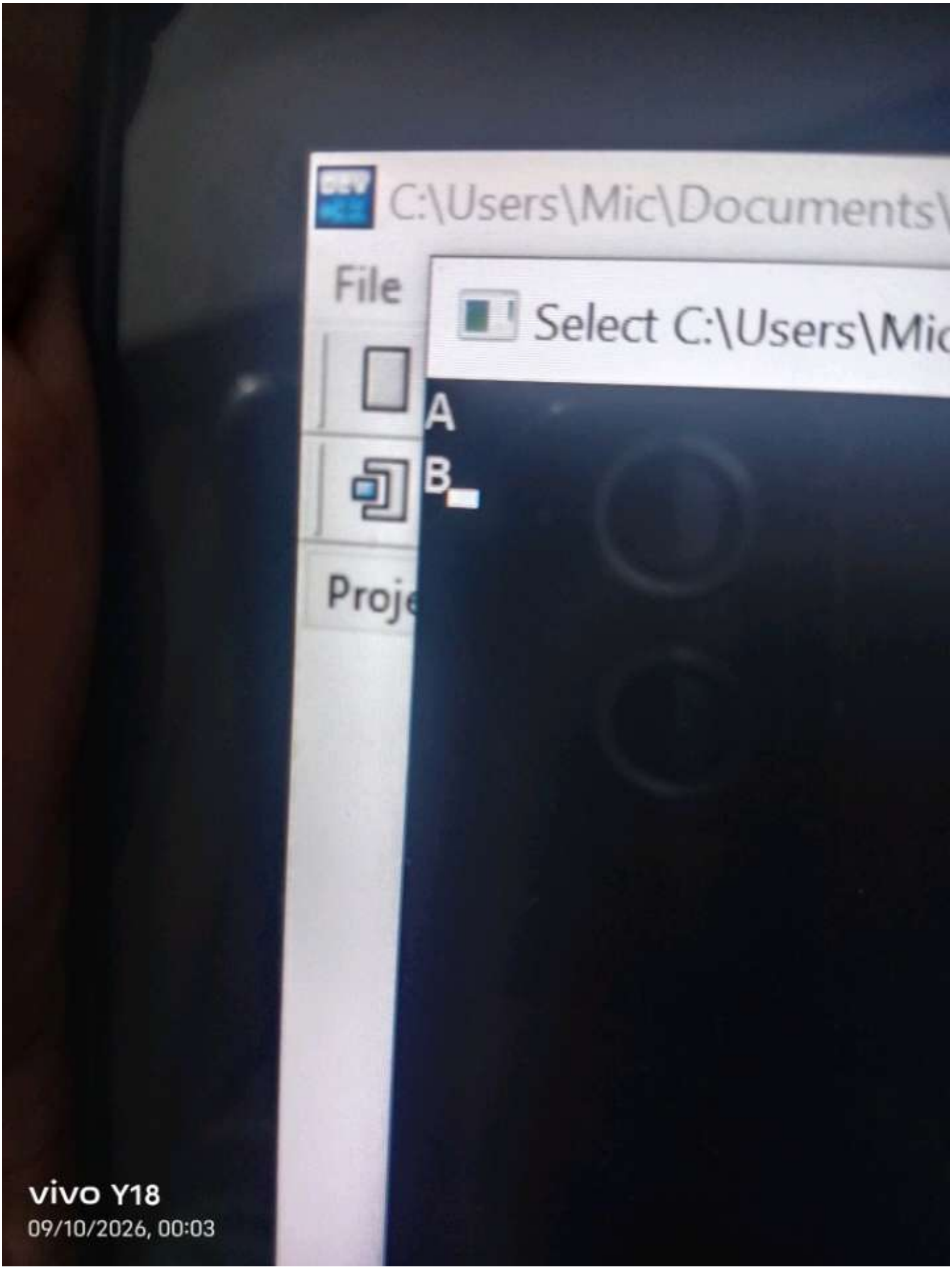
[*] TASK 1.cpp

```
1    TASK 1:
2    INPUT 2
3
4
5    # include <iostream>
6    using namespace std;
7    int main( ){
8    char letter;
9    letter= 65;
10   cout << letter<<'\n';
11   letter= 66;
12   cout << letter;
13   cin>> letter;
14   return 0;
15   }
```

Resources | Compile Log | Debug | Find Results | Cl

Compilation results...

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vivo Y18
09/10/2026, 00:03

[*] TASK 1.cpp

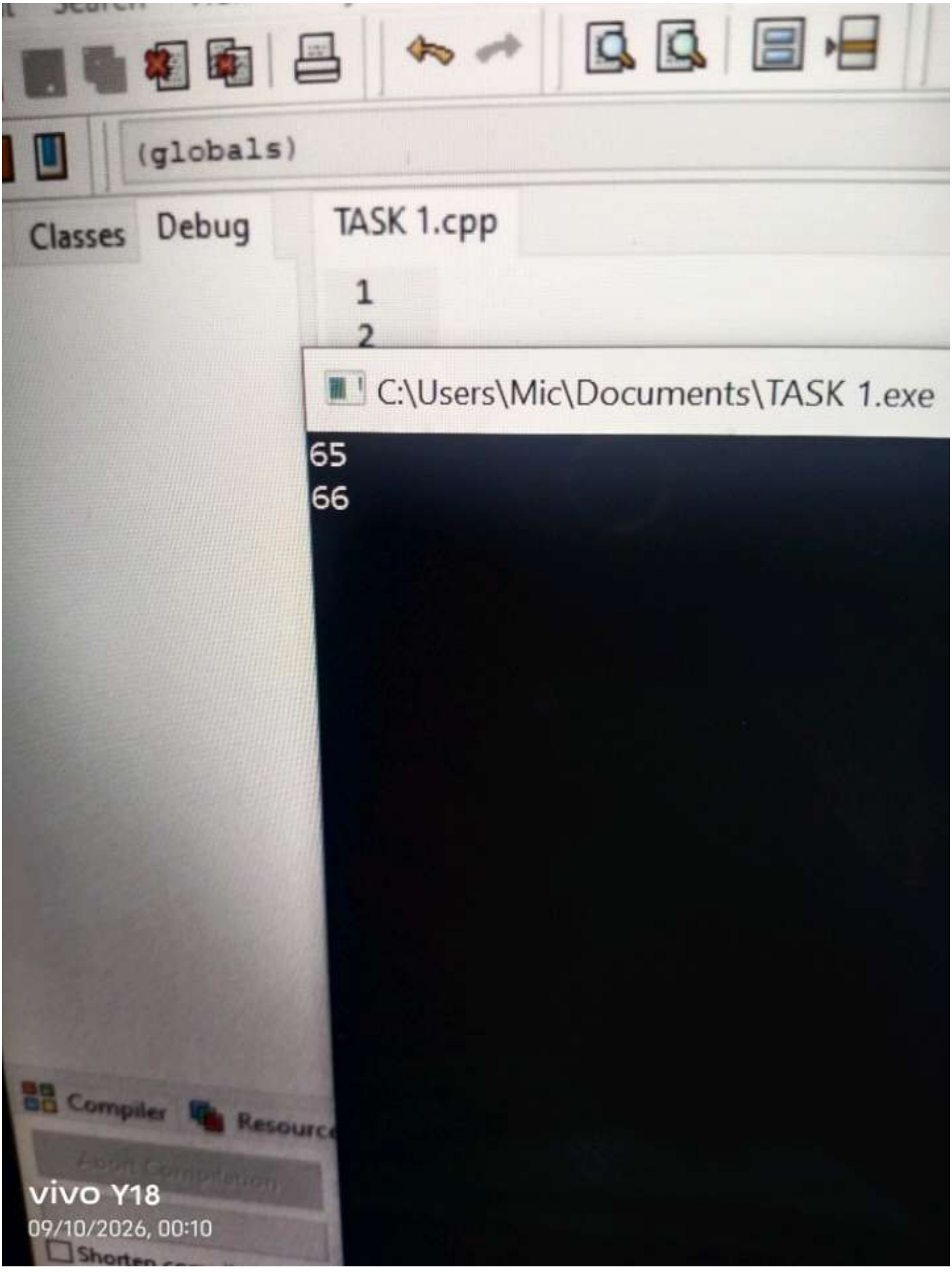
```
1   TASK 1:
2   INPUT 3
3
4
5   # include <iostream>
6   using namespace std;
7   int main( ){
8   int letter;
9   letter= 65;
10  cout << letter<<'\n';
11  letter= 66;
12  cout << letter;
13  cin>> letter;
14  return 0;
15  }
```

Resources | Compile Log | Debug | Find Results

Compilation results...

Errors: 0

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(globals)
Classes
Debug
TASK 1.cpp
1
2
C:\Users\Mic\Documents\TASK 1.exe
65
66
Compiler
vivo Y18
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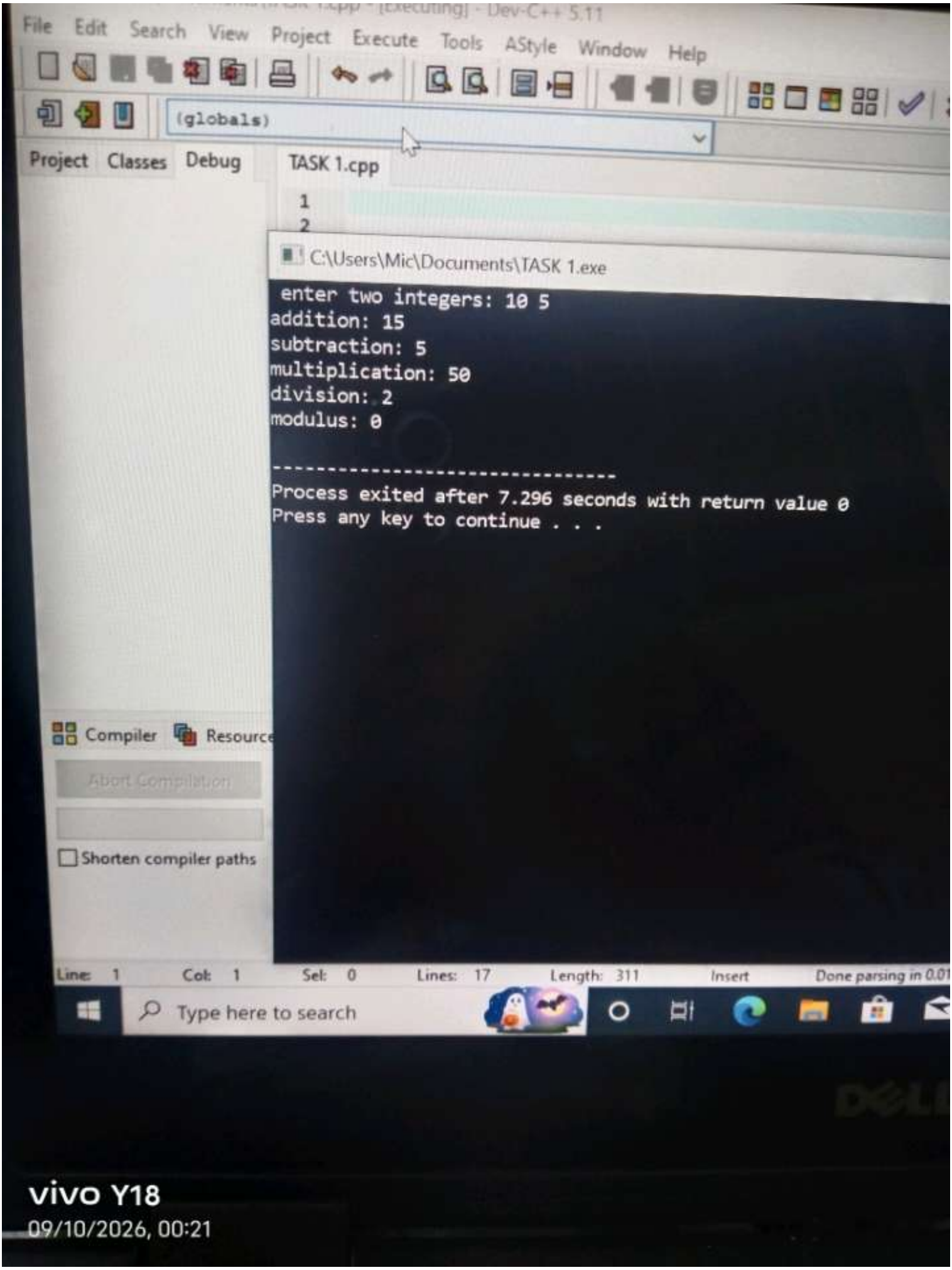
[*] TASK 1.cpp

```
1   TASK 2
2
3
4   # include <iostream>
5   using namespace std;
6   int main( ){
7
8   int a,b;
9   cout<<" enter two integers: ";
10  cin>> a>> b;
11  cout<<"addition: "<< a+b<<endl;
12  cout<<"subtraction: "<< a-b<<endl;
13  cout<<"multiplication: "<< a*b<<endl;
14  cout<<"division: "<< a/b<<endl;
15  cout<<"modulus: "<< a%b<<endl;
16  return 0;
17  }
```

Resources | Compile Log | Debug | Find Results | Close

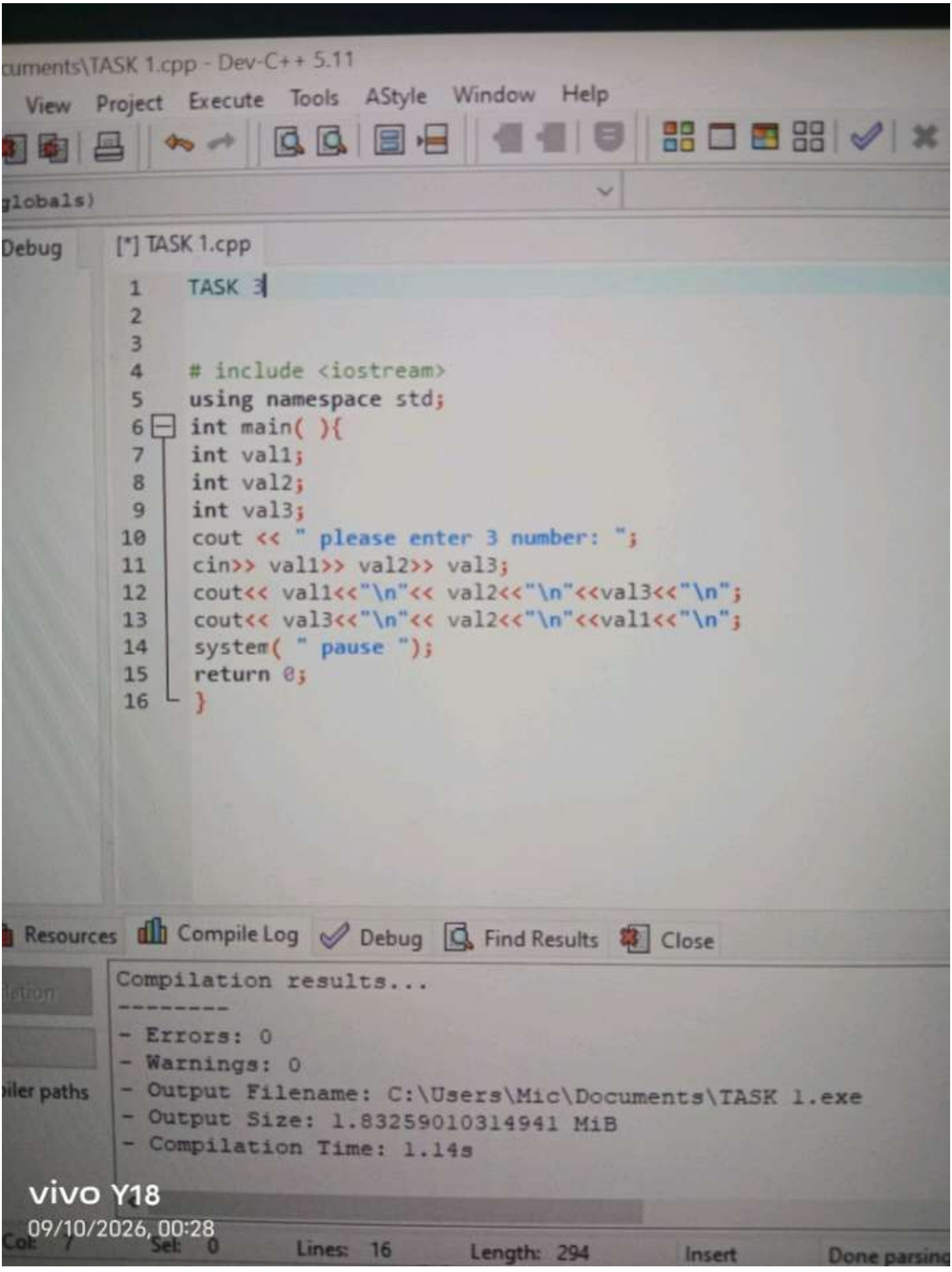
```
Compilation results...
--------
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\Mic\Documents\TASK
- Output Size: 1.833097457885 74 MiB
- Compilation Time: 1.00s
```

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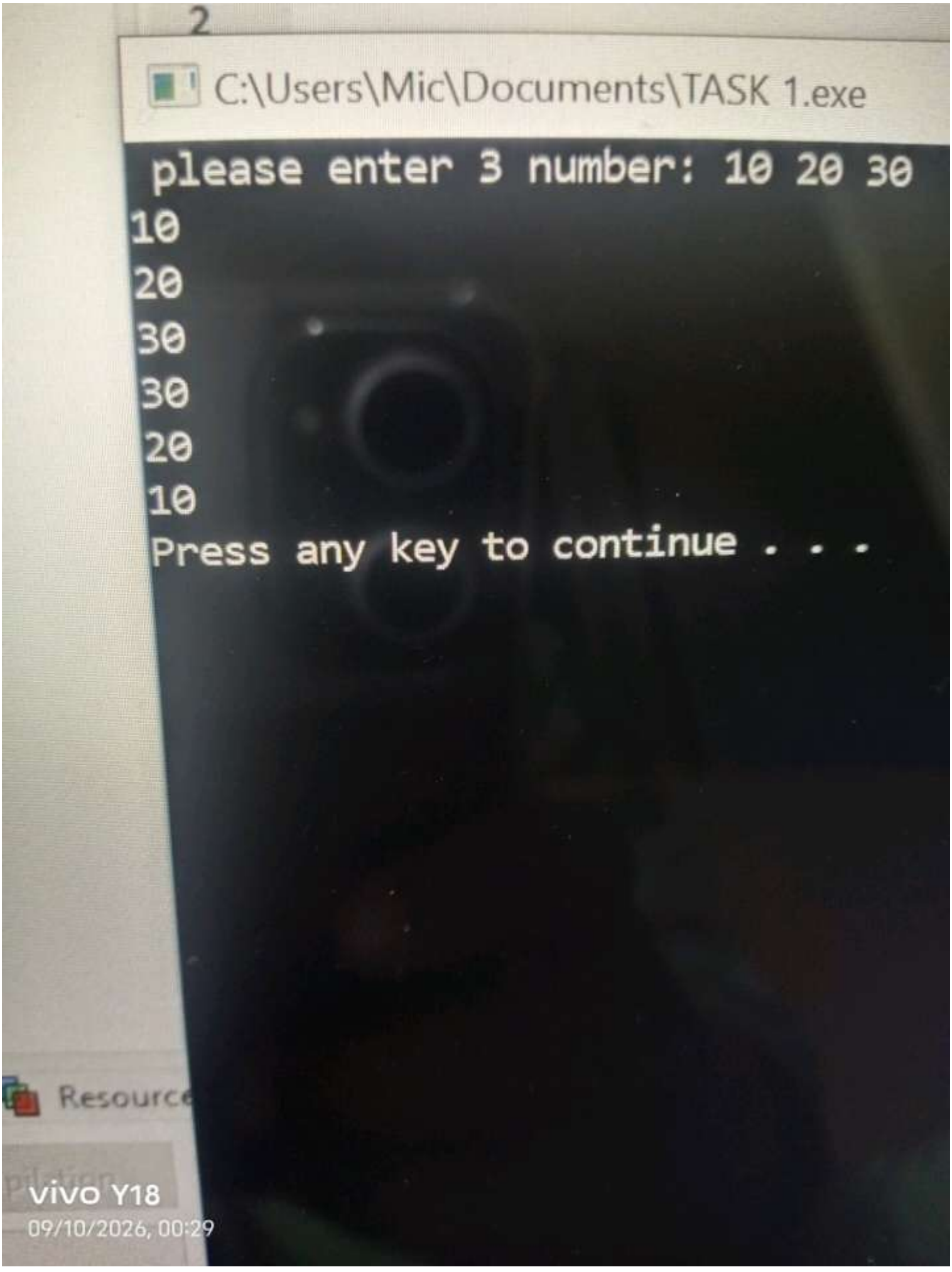


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2
C:\Users\Mic\Documents\TASK 1.exe
please enter 3 number: 10 20 30
10
20
30
30
20
10
Press any key to continue . . .
Resource
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TASK 1.cpp

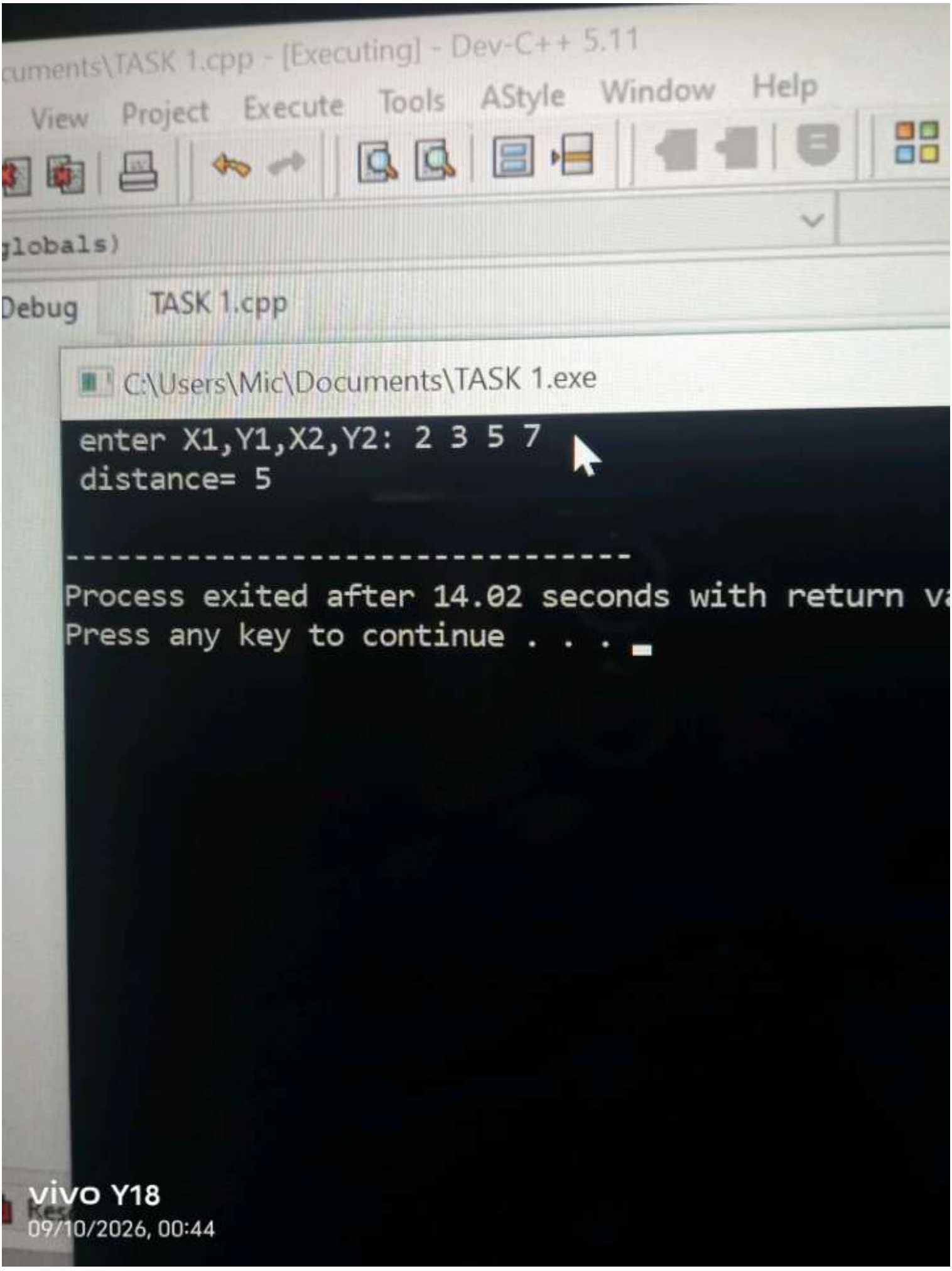
TASK 4

```cpp
# include <iostream>
# include <cmath>
using namespace std;
int main( ){
double X1,Y1,X2,Y2,d;
cout<< " enter X1,Y1,X2,Y2: ";
cin>>X1>>Y1>>X2>>Y2;
d=sqrt(pow( X2-X1,2 )+pow( Y2-Y1,2));
cout<<" distance= "<< d<< endl;
return 0;
}
```

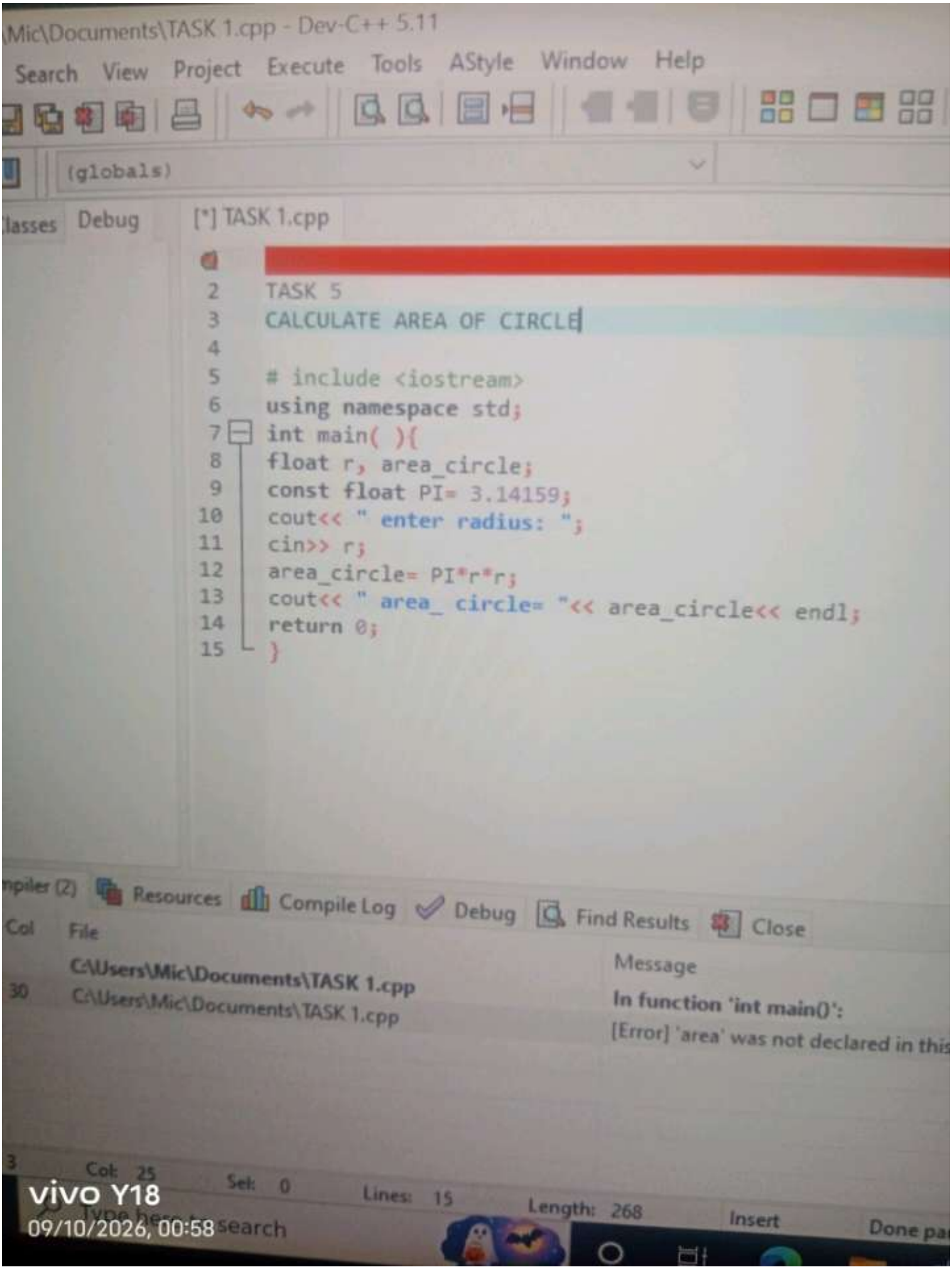
Compile Log | Debug | Find Results | Close

Compilation results...

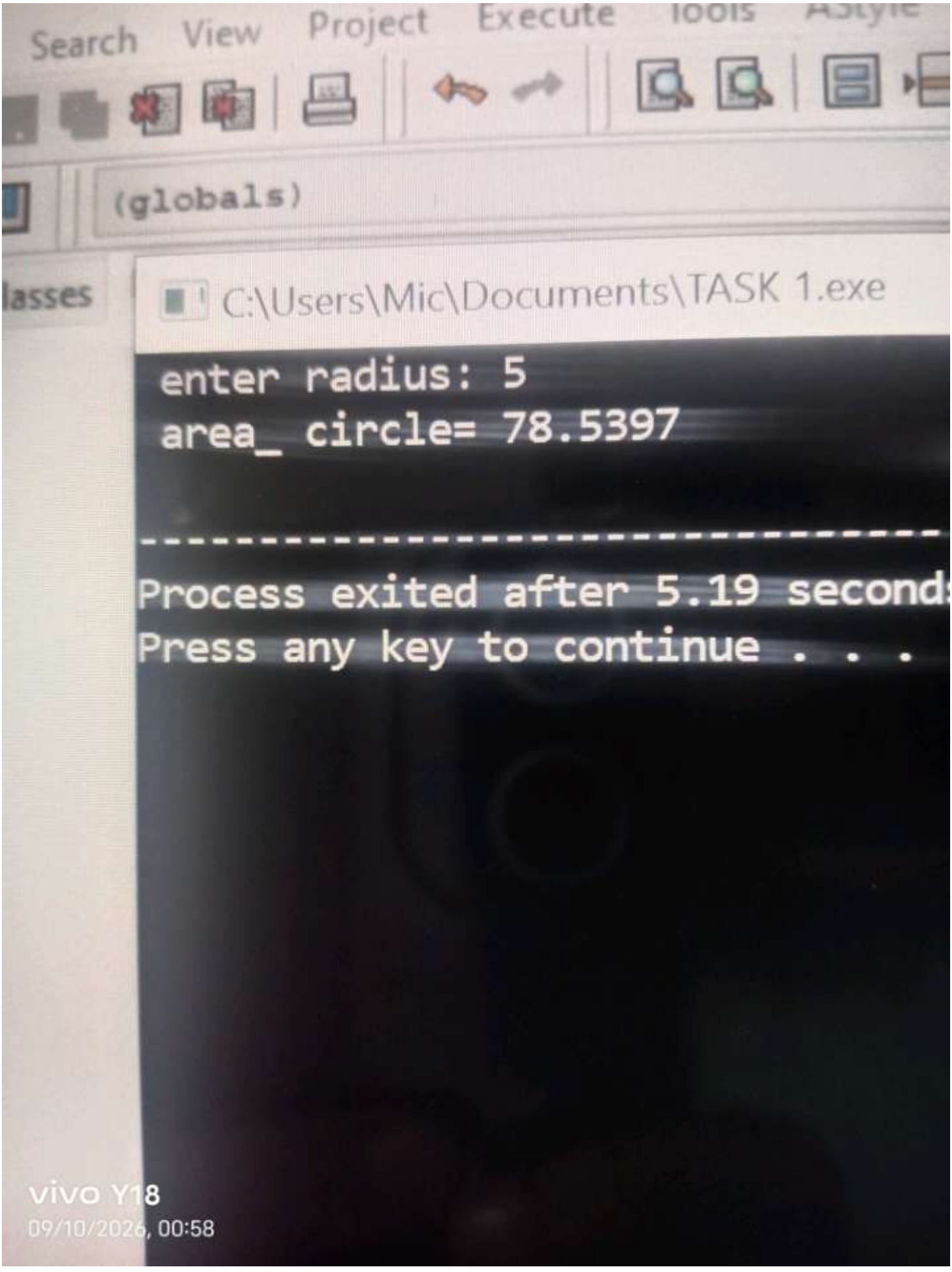
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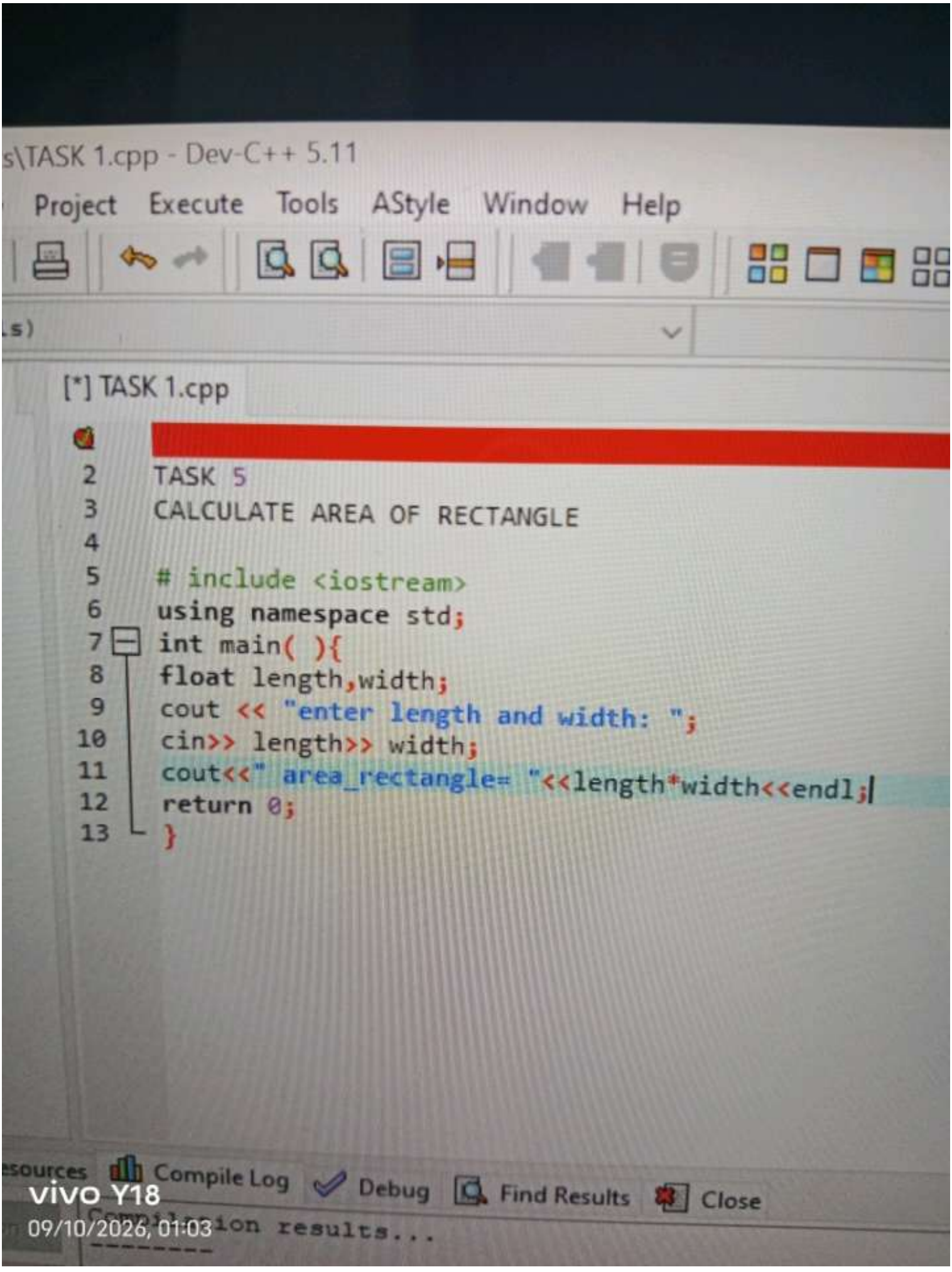


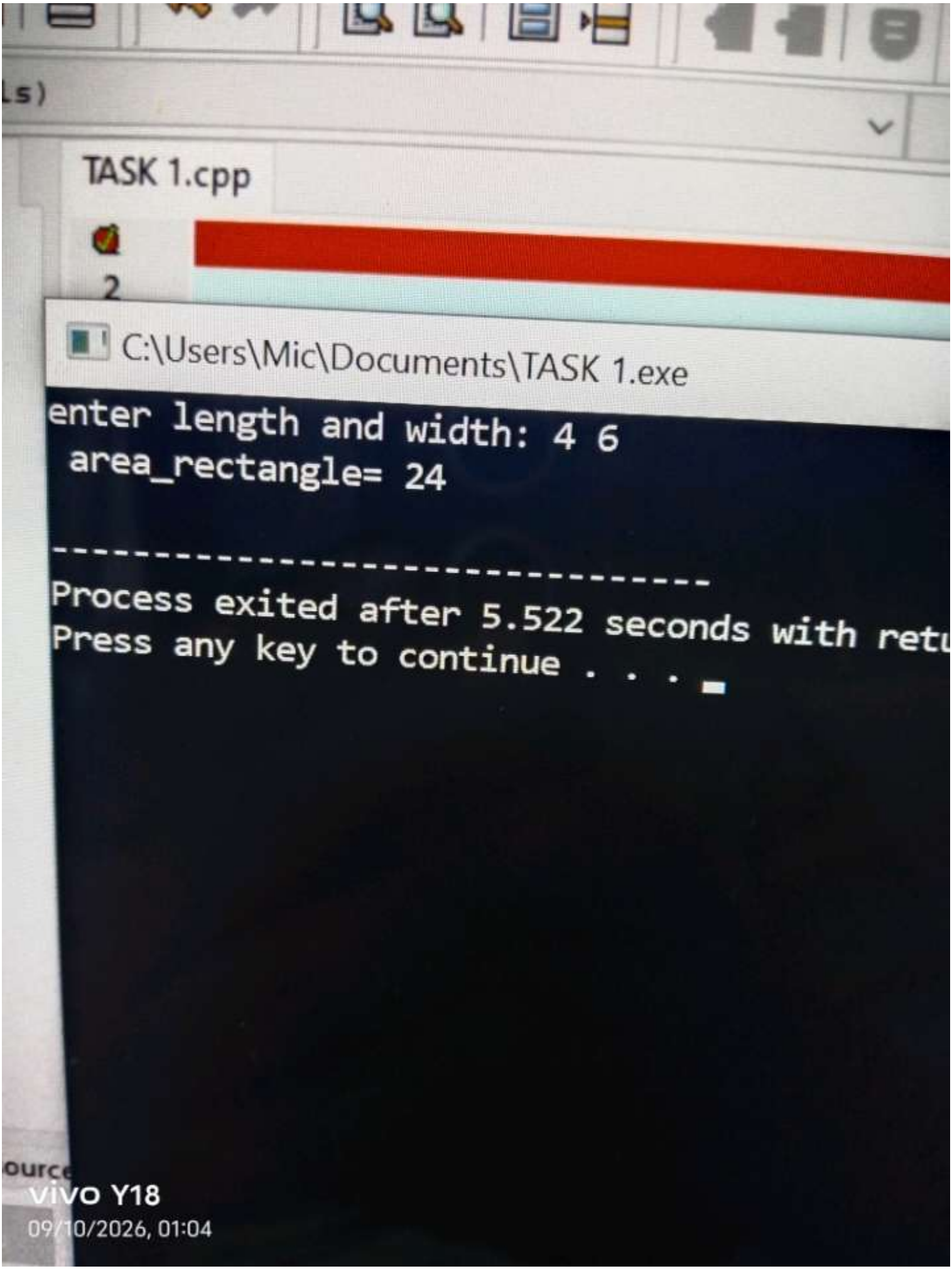
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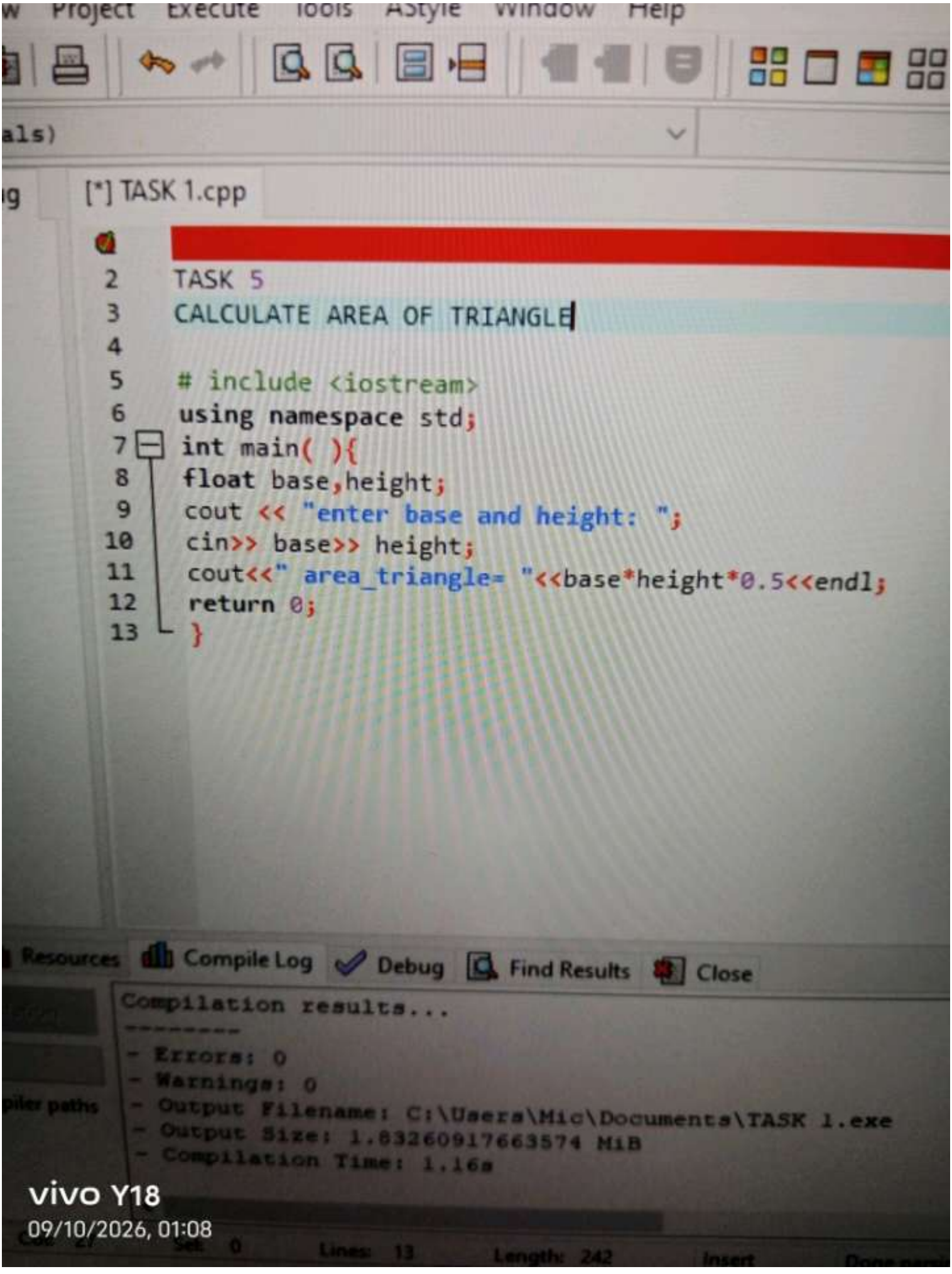
vivo Y18
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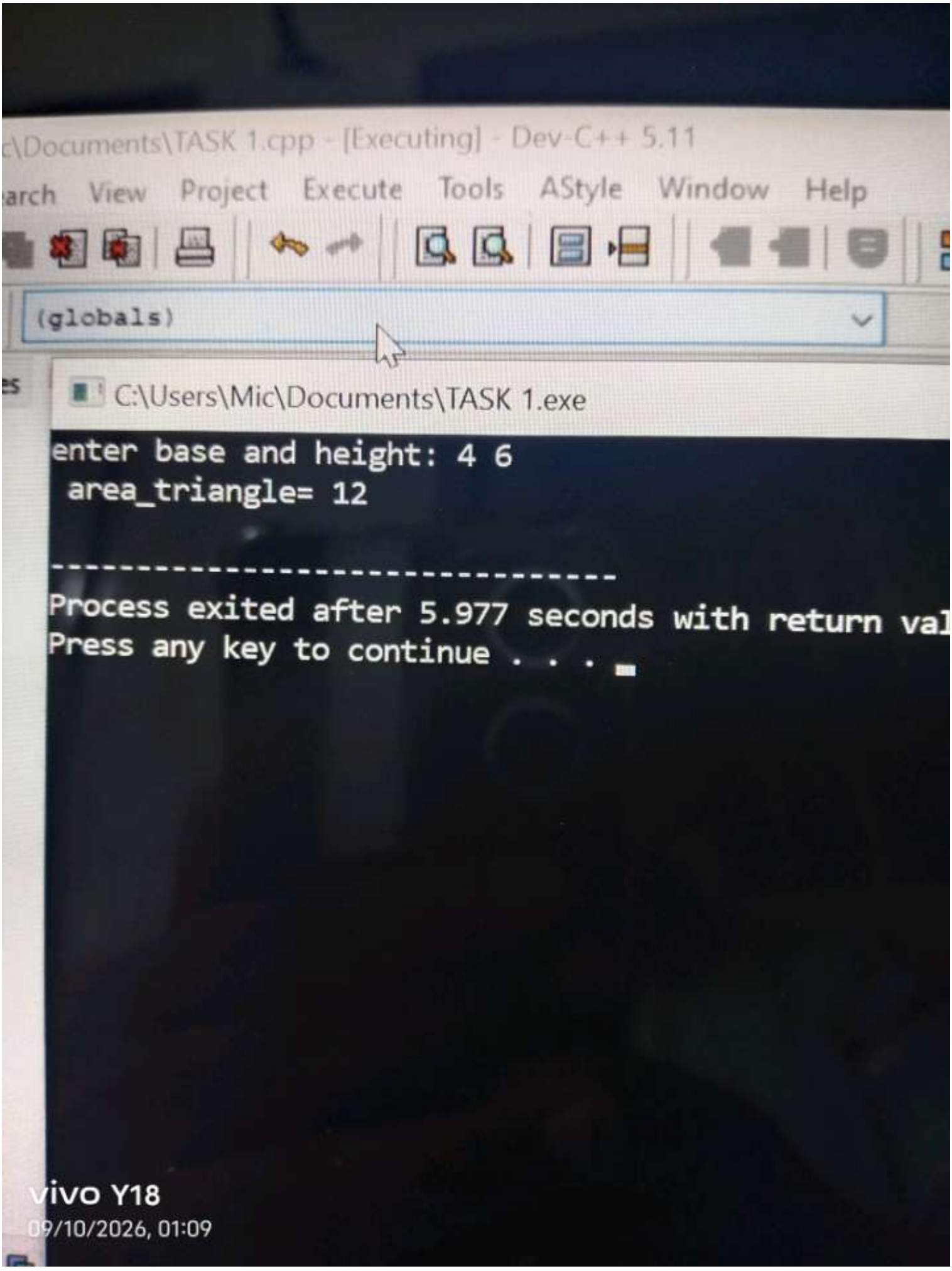

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(globals)
lasses
C:\Users\Mic\Documents\TASK 1.exe
enter radius: 5
area_ circle= 78.5397
Process exited after 5.19 seconds
Press any key to continue . . .
vivo Y18
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s\TASK 1.cpp - Dev-C++ 5.11
Project Execute Tools AStyle Window Help
[*] TASK 1.cpp
TASK 5
CALCULATE AREA OF RECTANGLE
include <iostream>
using namespace std;
int main( ){
float length,width;
cout << "enter length and width: ";
cin>> length>> width;
cout<<" area_rectangle= "<<length*width<<endl;
return 0;
}
Compile Log
Debug
Find Results
Close
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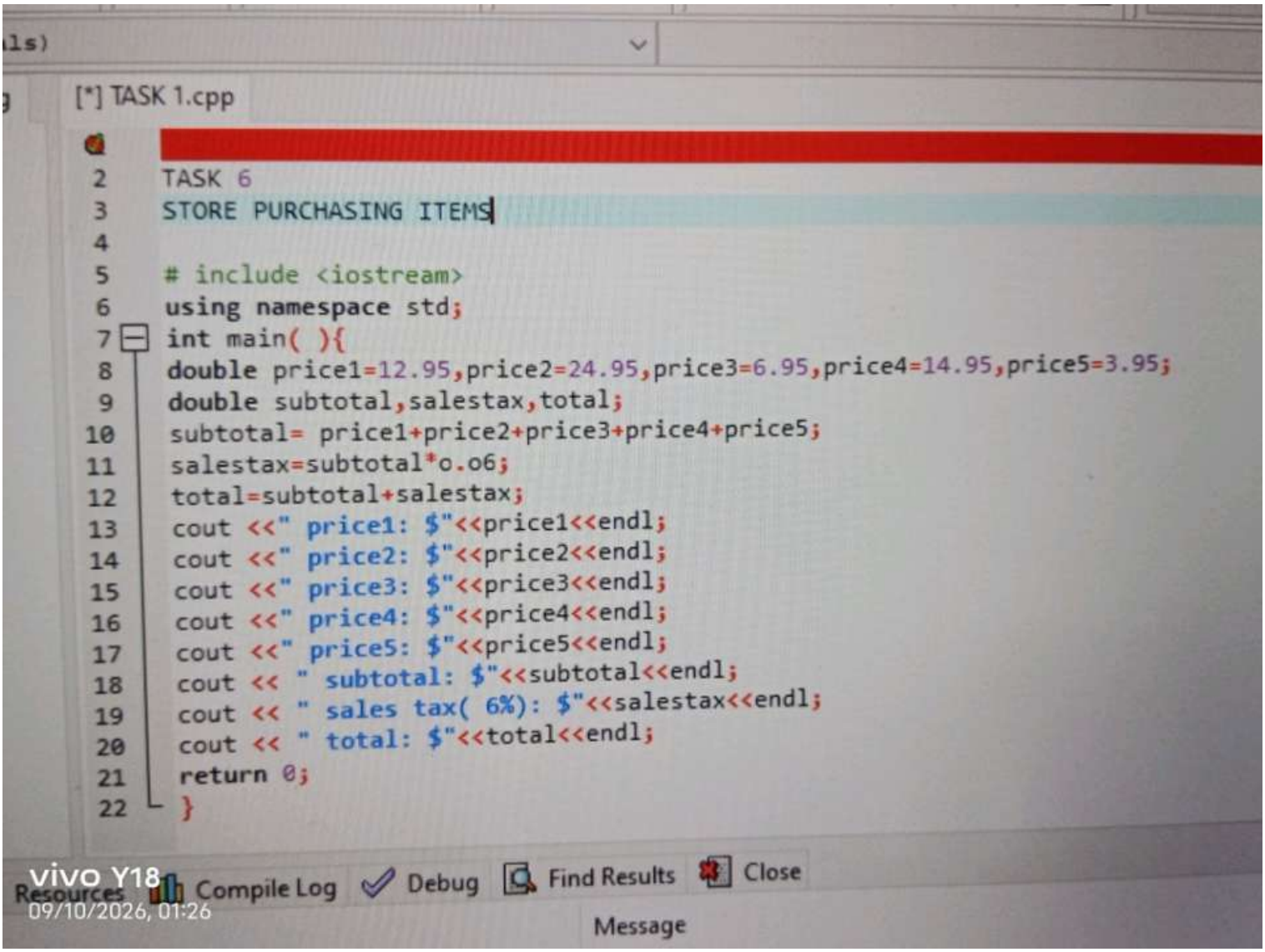


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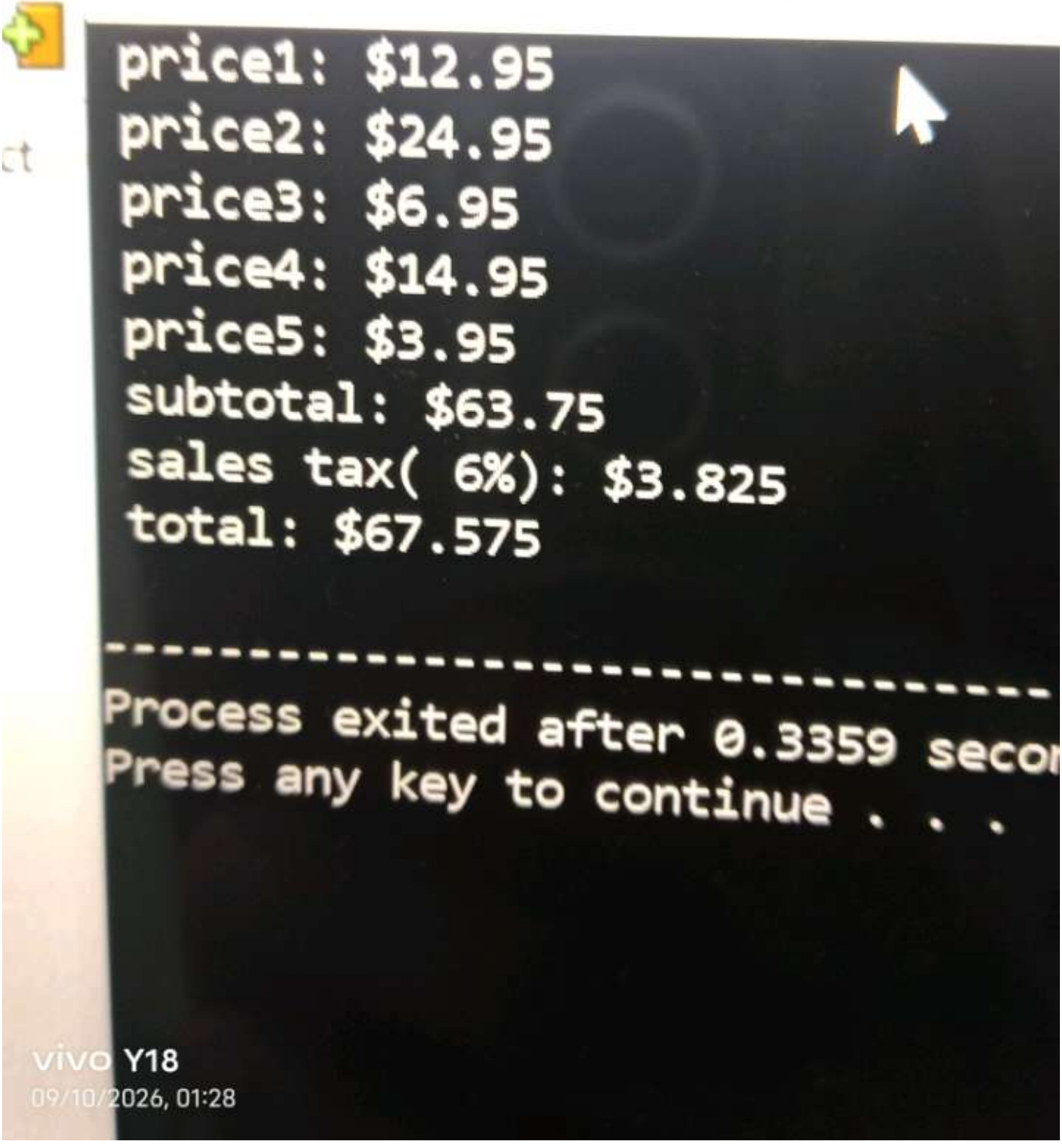

Project Execute Tools AStyle Window Help
[*] TASK 1.cpp
TASK 5
CALCULATE AREA OF TRIANGLE
include <iostream>
using namespace std;
int main( ){
float base,height;
cout << "enter base and height: ";
cin>> base>> height;
cout<<" area_triangle= "<<base*height*0.5<<endl;
return 0;
}
Resources
Compile Log
Debug
Find Results
Close
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\Mic\Documents\TASK 1.exe
- Output Size: 1.832609176635 74 MiB
- Compilation Time: 1.16s
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vivo Y18
09/10/2026, 01:09



vivo Y18
09/10/2026, 01:26



vivo Y18
09/10/2026, 01:28